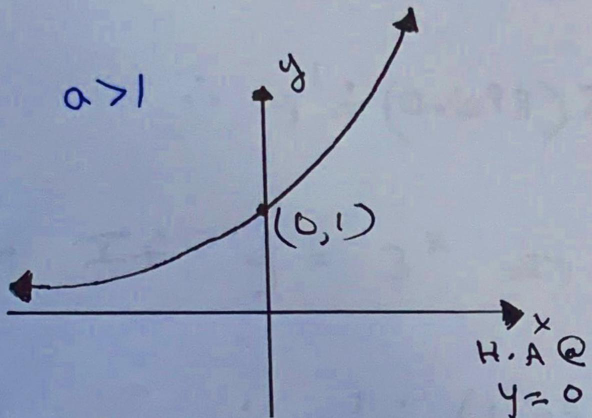
Exponential Functions

Exponential Function: $y = a^x \quad a \in \mathbb{R},$

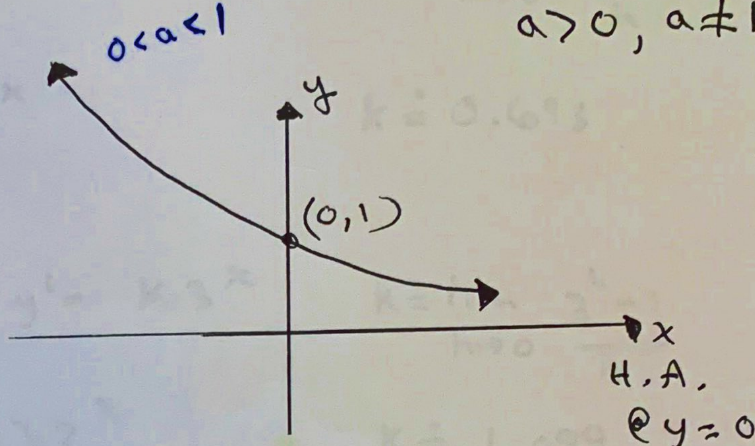
$a > 0,\ a \neq 1.$



Increasing function

Domain: $(-\infty, \infty)$

Range: $(0, \infty)$



Decreasing function

Domain: $(-\infty, \infty)$

Range: $(0, \infty)$

NOTE: The derivative of an exponential function is an exponential function.

Given $f(x) = a^x$

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{a^{x+h} - a^x}{h}$$

$$= a^x \cdot \underbrace{\lim_{h \to 0} \frac{a^h - 1}{h}}$$

exponential some constant

Consider $y = a^x$

* If $y = 2^x \Rightarrow y' = k \cdot 2^x$ $\qquad k = \lim_{h \to 0} \frac{2^h - 1}{h}$

$\therefore y' \doteq (0.693) 2^x$ $\qquad k \doteq 0.693$

* If $y = 3^x \Rightarrow y' = k \cdot 3^x$ $\qquad k = \lim_{h \to 0} \frac{3^h - 1}{h}$

$\therefore y' \doteq (1.099) \cdot 3^x$ $\qquad k \doteq 1.099$

* Question:

Is there a number between 2 and 3 such that $\lim_{h \to 0} \frac{a^h - 1}{h} = 1$?

If so, we have found a function which is its own derivative.

→ Answer: Yes!

"e" $\approx$ 2.71828... (named after Euler "oiler")

$\therefore \lim_{h \to 0} \frac{e^h - 1}{h} = 1$

$\therefore$ If $y = e^x \Rightarrow y' = e^x$

In general, if

$f(x) = e^{g(x)}$, then $f'(x) = e^{g(x)} \cdot g'(x)$

example 1: Differentiate the following.

a) $y = 2e^x$

$y' = 2e^x$

b) $y = e^{2x}$

$y' = e^{2x} \cdot (2x)'$

$y' = 2 \cdot e^{2x}$

c) $y = e^{-3x^2}$

$e^{-3x^2} \cdot (-6x)$

$y' = e^{-3x^2} (-3x^2)'$

$y' = -6x \cdot e^{-3x^2}$

d) $f(x) = 3x^2 \cdot e^x$

$f'(x) = 6x \cdot e^x + 3x^2 \cdot e^x$

$f'(x) = 3xe^x \cdot (2 + x)$

e) $f(x) = \dfrac{x^2}{e^{2x+1}}$

$$f'(x) = \frac{2x\,e^{2x+1} - x^2 \cdot e^{2x+1} \cdot (2)}{(e^{2x+1})^2}$$

$$f'(x) = \frac{2xe^{\cancel{2x+1}}(1-x)}{(e^{2x+1})^{\cancel{2}}} = \frac{2x\,(1-x)}{e^{2x+1}}$$

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f) $f(x) = e^{x^2} \cdot \sin(x^2)$

$f'(x) = \left(e^{x^2}\right)' \sin(x^2) + e^{x^2}\left(\sin(x^2)\right)'$

$f'(x) = 2xe^{x^2} \sin(x^2) + e^{x^2} \cos(x^2) \cdot 2x$

$f'(x) = 2xe^{x^2}\left(\sin(x^2) + \cos(x^2)\right)$

$\sin(\theta) + \cos(\theta) = 0$

$\sin(\theta) = -\cos(\theta)$

$\tan(\theta) = -1$

$[0, \pi]$

$\theta = \tan^{-1}(-1)$

$\theta = \frac{3\pi}{4}$

$x^2 = \frac{3\pi}{4}$

$x =$

g) $f(x) = \underbrace{\pi e^{\pi}}\cdot \sin(e^x) + e^{\pi}$ ← constant

$f'(x) = \pi e^{\pi} \cdot \cos(e^x) \cdot e^x + 0$

$f'(x) = \pi \cdot e^{x+\pi} \cdot \cos(e^x)$

$e^{\pi} \rightarrow 0$

$e^{\sqrt{\pi}} \rightarrow 0$

$e^{\pi - 1} \rightarrow 0$

h) Determine the equation of the tangent to the curve $y = e^{2x-1}$ at $x = 1$.

$\rightarrow y = e^{2(1)-1}$

$y = e$.

$y = e^{2x-1}$

$y' = 2 \cdot e^{2x-1}$

$m_t\big|_{x=1} = 2 \cdot e$ ← leave exact

$y - y_1 = m(x - x_1)$

$y - e = 2e(x-1)$ ← p.s. form.

$y - e = 2ex - 2e$

$y = \underbrace{2e}_{m} x - e$ slope-y-int

← y-int

$Ax + By + C = 0$

$\longrightarrow 2ex - y - e = 0$